Logistics Performance & Sales Analysis Report

Objective

The main objective of this project is to analyze logistics and supply chain data to gain insights into revenue, profitability, delivery performance, geographical distribution, and customer satisfaction. The dashboard helps stakeholders understand operational trends, identify inefficiencies, optimize costs, and improve decision-making through data-driven insights.

Purpose

To provide a business intelligence dashboard that enables:

- Monitoring of key financial metrics such as revenue, profit, and shipping costs
- Tracking delivery performance including on-time and delayed shipments
- Analyzing geographical shipment flow and carrier efficiency
- Evaluating customer experience through ratings and delivery timelines
- Supporting strategic decisions to improve operational efficiency and profitability

Dataset Summary

The dataset used in this project represents logistics and supply chain operations, containing detailed information about orders, shipments, customers, and financial performance.

- Total Records – Approximately 75000 details.
- Total columns – 22 Attributes after data cleaning and Normalization.
- Type of Data – Logistics performance & sales, Time based, categorical, and Numerical
- Format -CSV imported in to PostgreSQL database for Normalization.
- Source – Synthetic dataset prepared for analytical academic Use.

Key cleaning and Transformation steps:

- Removed Duplicates – Duplicate records were identified and removed to maintain data integrity and avoid incorrect aggregations.
- Handled Missing Values – Null values in date fields (such as order and delivery dates) were replaced or corrected to enable proper time-based analysis.
- Trimmed & Cleaned Text Fields – Extra spaces, hidden characters, and inconsistencies in text columns were removed using trim and clean functions.

- Standardized Text Format – Text values were formatted using proper capitalization to ensure uniformity across categorical fields.
- Corrected Spelling Errors – Inconsistent or incorrect text entries (such as region names, shipment status, and categories) were standardized to improve data quality.
- Validated Data Types – Columns were assigned appropriate data types (e.g., date, numeric, text) to support accurate calculations and visualizations.
- Created Derived Columns – Additional fields such as *Delivery Status*, *Delivery Days*, and *Shipping Mode* were created to enhance analysis.
- Ensured Data Consistency – Relationships between tables were validated to support the star schema model and ensure proper filtering across visuals.

Data model overview

The project follows a star schema model to optimize performance and ensure data scalability.

Central table (fact table)

fact_orders
Contains the main transactional data for each order, including revenue, profit, shipping cost, dates, and customer details. It is the central table used for analysis.

Dimention Tables

dim_product
Stores product-related information such as product category and product identifiers, used to analyze performance by product type.

- **dim_geography**
  Contains location details like origin and destination regions, used to analyze shipment flow and regional performance.
- **dim_shipping**
  Includes shipment-related attributes such as shipping mode, carrier type, and shipment status, used to evaluate delivery performance.
- **Dim_Date**
  Provides date-related information such as day, month, and year, used for time-based analysis and trend tracking.

Relationships

The data model follows a star schema, where the fact_orders table is connected to dimension tables using key relationships.

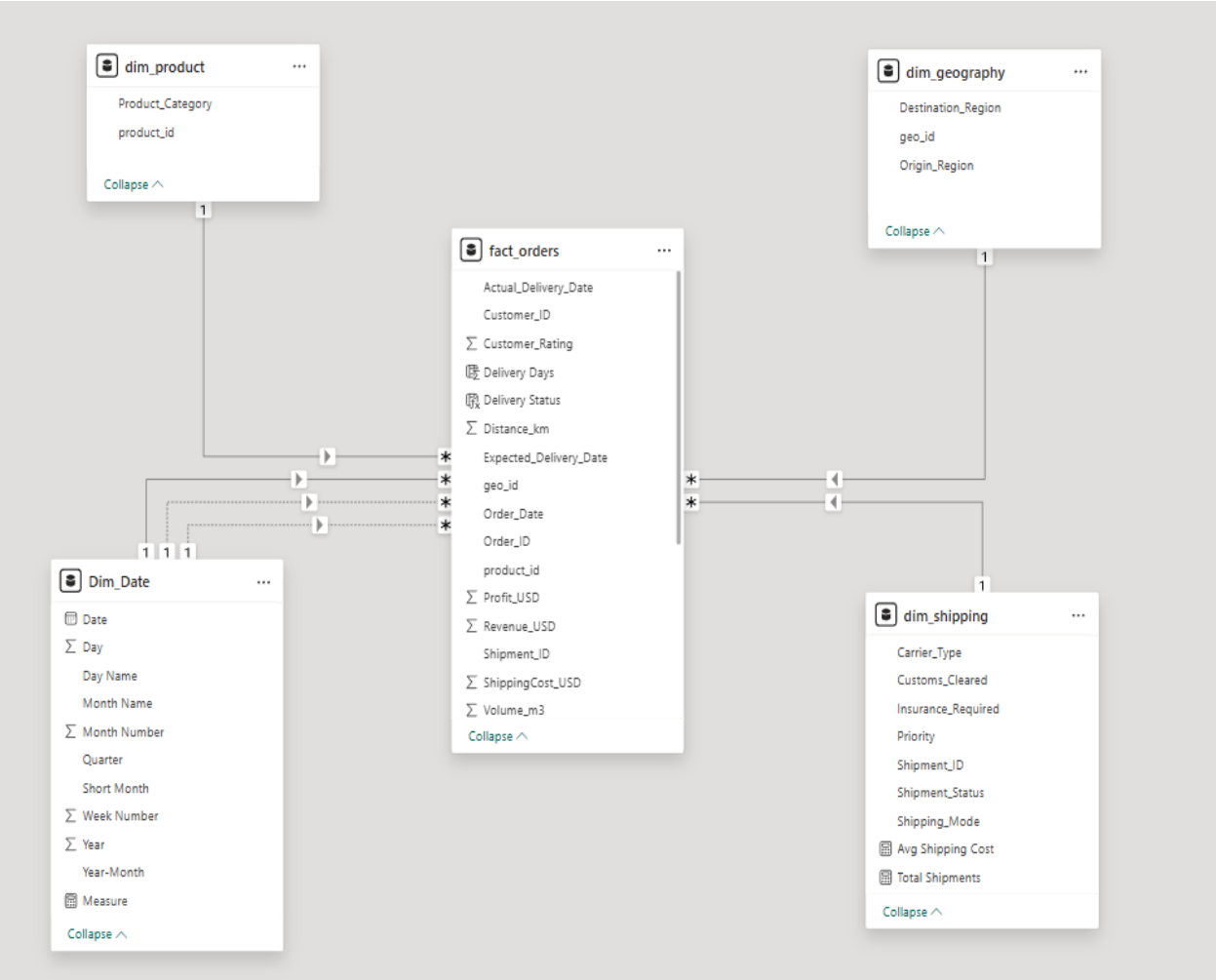
- Dimension tables contain Primary Keys (e.g., product_id, geo_id, Shipment_ID, Date)
- The fact_orders table contains corresponding Foreign Keys
- Relationships are one-to-many (1 → *), where:
  - One record in dimension → many records in fact table

This structure enables accurate filtering, efficient data analysis, and improved performance.

Calculated Fields & Measures

- **Calculated Fields (Columns)** were created to derive additional insights from existing data, such as:
  - *Delivery Status* (based on shipment status)
  - *Delivery Days* (difference between order and delivery dates)
  - *Shipping Mode* (retrieved from shipping dimension)
- **Measures** were created using DAX to perform dynamic calculations, including:
  - *Total Revenue, Total Profit, Total Orders*
  - *Profit Margin (%)*
  - *On-Time and Delayed Deliveries*
  - *Delay Rate (%), Average Rating, and Average Shipping Cost*

These calculations enable interactive analysis and support data-driven decision-making in the dashboard.



1. Sales Performance:

- Total revenue shows a steady growth trend over time.
- Profit margins remain stable across most periods.
- Peak sales are observed during specific months/seasonal periods.

2. Logistics Performance:

- Most orders are delivered successfully with a high delivery rate.
- Delays are more common in certain regions and shipment modes.
- Average delivery time varies based on shipping method.

3. Regional Performance:

- Major regions contribute the highest revenue and order volume.
- Some smaller regions show lower orders but higher profit margins.

4. Product Performance:

- Top products contribute significantly to total revenue.
- Few products show low performance and need improvement.

5. Customer Insights:

- A small group of customers contributes a large portion of revenue.
- Repeat customers play a key role in overall sales growth.

Conclusion

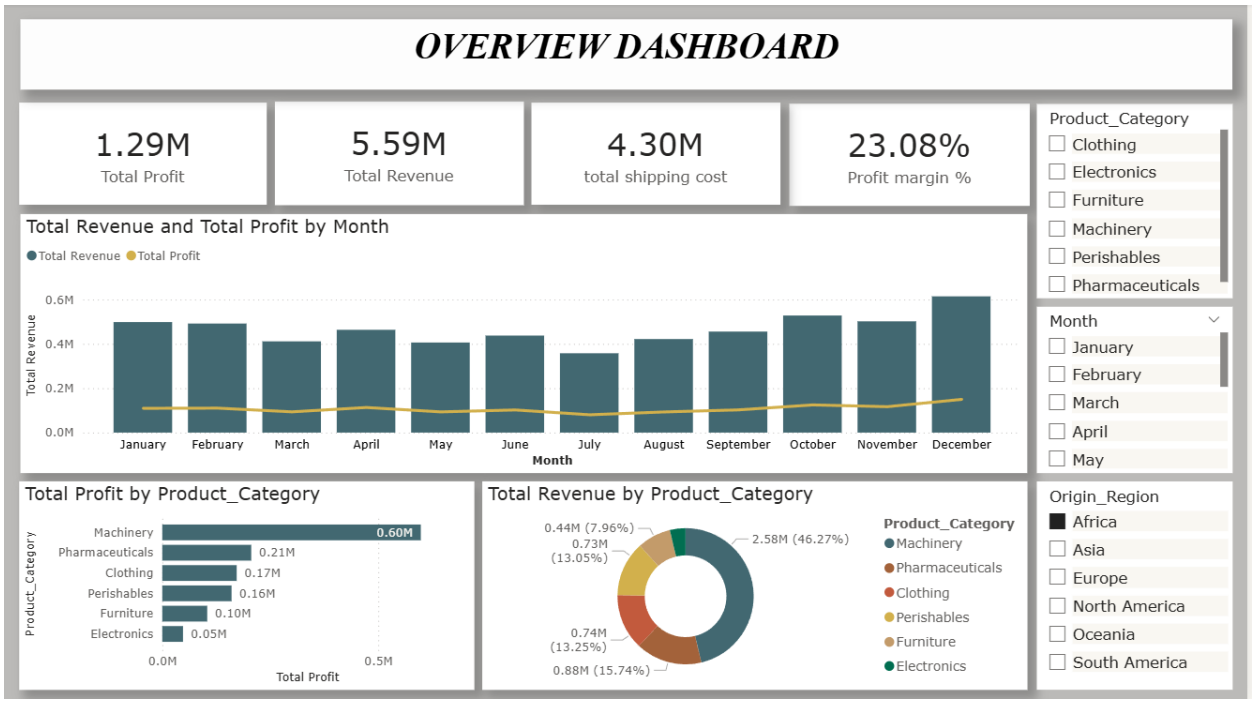
- The dashboard provides a clear overview of sales, logistics, and customer performance.
- Revenue and order trends show consistent business growth.
- Delivery performance is strong, with most orders completed successfully.
- Data-driven insights help in identifying high-performing regions, products, and customers.

Recommendations

- Improve delivery efficiency in regions with higher delays.
- Focus on promoting low-performing products to increase sales.
- Strengthen relationships with high-value and repeat customers.
- Optimize shipping methods to reduce delivery time and cost.
- Use insights for better demand forecasting and inventory planning.

Dashboard Overview

Overview Dashboard



Purpose

Summarizes financial performance (profit, revenue, shipping cost, margin) to guide business decisions.

Visuals Used

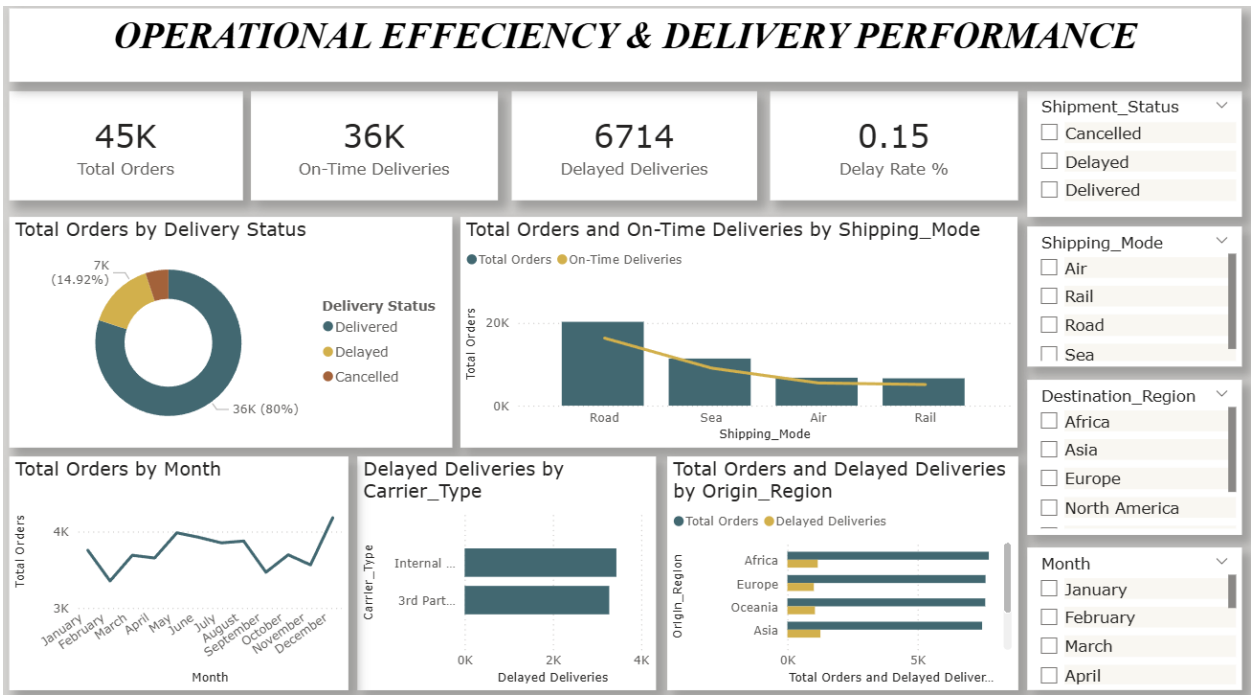
- **KPIs:** Profit, revenue, shipping cost, margin %.
- **Bar + Line Chart:** Revenue vs. profit by month.
- **Bar Chart:** Profit by product category.
- **Donut Chart:** Revenue share by product category.
- **Filters:** Category, month, origin region

Insights Provided

- **Profitability:** 1.29M profit, margin ≈23%.

- **Category Performance:** Machinery leads in both profit and revenue.
- **Revenue Trend:** Monthly revenue and profit fluctuate, peaking in some months.
- **Distribution:** Revenue concentrated in Machinery, Pharmaceuticals, and Perishables.

Operational Efficiency & Delivery Performance



Purpose

Measures order handling, delivery timeliness, and carrier/region efficiency to improve operations.

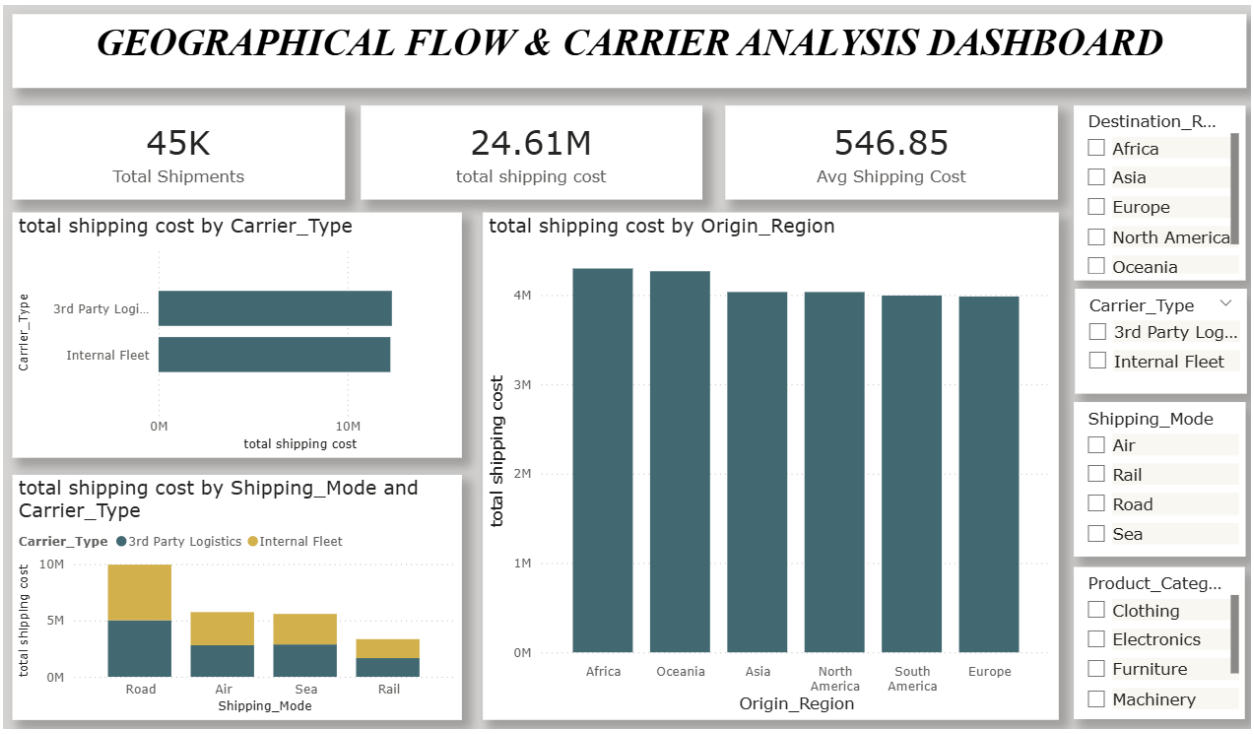
Visuals Used

- **KPIs:** Total orders, on-time, delayed, delay rate.
- **Pie Chart:** Orders by delivery status.
- **Bar + Line Chart:** Orders vs. on-time by shipping mode.
- **Line Chart:** Orders trend by month.
- **Bar Charts:** Delays by carrier type, orders vs. delays by region.
- **Filters:** Shipment status, mode, region, month.

Insights Provided

- **Delivery Performance:** 80% delivered on time, delay rate very low (0.15%).
- **Carrier Analysis:** 3rd Party vs Internal delays compared.
- **Regional View:** Some regions show higher delays.
- **Seasonal Trend:** Orders peak in December.

Geographical Flow & Carrier Analysis Dashboard



Purpose

This dashboard is designed to analyze shipping performance and costs across regions, carrier types, and shipping modes. It helps supply chain managers identify cost drivers, compare logistics strategies, and optimize carrier usage for efficiency and savings.

Visuals Used

- KPIs: Total shipments, total cost, avg cost.
- Bar Charts: Costs by carrier type and origin region.
- Clustered Bar Chart: Costs by shipping mode & carrier type.
- Filters: Destination, carrier, mode, product category.

Insights Provided

- **Carrier Costs:** Compare 3rd Party Logistics vs Internal Fleet.
- **Regional Costs:** Africa shows highest shipping expenses.
- **Mode Analysis:** Costs vary across Air, Road, Sea, Rail.
- **Overall Benchmark:** 45K shipments, avg cost ≈547 per shipment.

Customer Experience & Analysis Dashboard

Purpose

A Customer Experience & Quality Dashboard is built to track and evaluate customer satisfaction, order volumes, and delivery performance. It helps businesses identify trends, strengths, and areas needing improvement.

Visuals Used

- **KPIs (Top Metrics):** Quick snapshot of totals (customers, orders, ratings, delivery days).
- **Bar Charts:** Compare average ratings across product categories and regions.
- **Line Chart:** Show delivery performance trends over months.
- **Donut Chart:** Display distribution of orders by customer rating.

- **Filters:** Allow interactive drill-down by category, region, month, and rating.

Insights Provided

- **High Satisfaction:** Most orders come from customers rating 4 or 5, showing strong overall experience.
- **Delivery Trend:** Average delivery days are decreasing, indicating improved efficiency.
- **Regional Variation:** Some regions score higher in ratings, highlighting geographic differences.
- **Category Performance:** Certain product categories perform better than others, guiding focus for improvement.

Notes / Limitations

- Focused but siloed: Each dashboard covers one aspect (customer, logistics, operations, finance) but they don't connect insights across areas.
- Averages mask detail: Metrics like average delivery days or costs may hide extremes and variations.
- Missing impact links: Customer ratings, delays, and costs aren't tied directly to financial outcomes or profitability.
- External factors excluded: Market demand, tariffs, promotions, or seasonal influences aren't shown, which limits context.
- Great overview, limited depth: